

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data to Delta Tables

Title: Load Customer and Order Data to Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.

- The Delta tables for customer and order data exist or can be created.

Main Flow / Functional Steps:

1. Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and load it into a Delta table named "customer".
2. Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` and load it into a Delta table named "order".
3. The customer Delta table should have the schema: CustId, Name, EmailId, Region.
4. The order Delta table should have the schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

FR-DLT-002: Transform Order Data by Adding TotalAmount Column

Title: Transform Order Data by Adding TotalAmount Column

Description: The system must add a new column "TotalAmount" to the order Delta table, which is the product of "PricePerUnit" and "Qty" columns.

Preconditions:

The order Delta table exists and has the required columns: PricePerUnit and Qty.

Main Flow / Functional Steps:

1. Add a new column "TotalAmount" to the order Delta table.
2. Calculate the value of "TotalAmount" by multiplying "PricePerUnit" and "Qty" columns.

FR-DLT-003: Remove Null and Duplicate Records from Customer and Order Tables

Title: Remove Null and Duplicate Records from Customer and Order Tables

Description: The system must remove null and duplicate records from both customer and order Delta tables.

Preconditions:

- The customer and order Delta tables exist.

Main Flow / Functional Steps:

1. Remove null records from the customer Delta table.
2. Remove duplicate records from the customer Delta table.
3. Remove null records from the order Delta table.
4. Remove duplicate records from the order Delta table.

FR-DLT-004: Create ordersummary Table and Load Joined Customer and Order Data

Title: Create ordersummary Table and Load Joined Customer and Order Data

Description: The system must create an "ordersummary" table if it does not exist and load joined customer and order data into it using SCD type 2.

Preconditions:

- The customer and order Delta tables exist and have the required data.
- The "ordersummary" table does not exist or can be created.

Main Flow / Functional Steps:

1. Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
2. Join the customer and order Delta tables using the "CustId" field.
3. Load the joined data into the "ordersummary" table using SCD type 2.
4. The "ordersummary" table should have the schema: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.

FR-DLT-005: Update ordersummary Table on Customer Data Change

Title: Update ordersummary Table on Customer Data Change

Description: The system must update the "ordersummary" table whenever there is a change in the customer table, maintaining SCD type 2 history.

Preconditions:

- The "ordersummary" table exists and has the required data.
- The customer Delta table exists and has the required data.

Main Flow / Functional Steps:

1. Detect changes in the customer Delta table.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update the StartDate and EndDate columns accordingly to maintain history.

FR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

Title: Create customeraggregatespend Table and Load Aggregated Data

Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data from the "ordersummary" table.

Preconditions:

- The "ordersummary" table exists and has the required data.
- The "customeraggregatespend" table does not exist or can be created.

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
2. Aggregate the "TotalAmount" column from the "ordersummary" table, grouping by "Name" and "Date" columns.
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-007: Implement Data Processing using Delta Live Tables

Title: Implement Data Processing using Delta Live Tables

Description: The system must implement the data processing pipeline using Delta Live Tables.

Preconditions:

- All previous requirements have been implemented.

Main Flow / Functional Steps:

1. Implement the data processing pipeline using Delta Live Tables, covering all the previous requirements.